

DecodeLabs

Phishing Awareness Analysis

Mariam Ahmed
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Project 3: Phishing Awareness Analysis

Executive Summary

Phishing is one of the most common cyberattacks used by attackers to steal sensitive information such as usernames, passwords, banking details, and personal data. Attackers often impersonate trusted organizations and use social engineering techniques to manipulate victims into taking unsafe actions.

This report analyzes a suspicious email and identifies indicators that reveal it as a phishing attempt. The analysis follows cybersecurity awareness principles recommended by organizations such as the National Institute of Standards and Technology (NIST) and the Cybersecurity and Infrastructure Security Agency (CISA).

Sample Email

Sender:

security-update@paypa1-support.com

Subject:

URGENT: Verify Your Account Immediately

Message:

Dear Customer,

We detected unusual activity in your account. To avoid permanent suspension, please verify your account within 24 hours by clicking the link below:

<http://paypal-security-verification-login.com>

Failure to verify your account may result in account closure.

Thank you,

Security Team

Phishing Analysis

1. Suspicious Sender Address

The sender's email address appears to imitate a legitimate company. The attacker replaced the letter "l" in PayPal with the number "1", making the address look authentic at first glance.

Risk:

Users may trust the email because it resembles an official sender.

2. Suspicious URL

The provided URL is not an official PayPal domain.

Malicious Link:

<http://paypal-security-verification-login.com>

Legitimate Domain Example:

<https://www.paypal.com>

Risk:

The link may redirect users to a fake website designed to steal login credentials.

3. Sense of Urgency

The email states that the account will be suspended within 24 hours.

Risk:

Attackers create panic to prevent users from carefully examining the message before acting.

4. Generic Greeting

The message begins with:

Dear Customer

Legitimate organizations typically address customers using their registered names.

Risk:

Generic greetings indicate that the same message is being sent to many potential victims.

5. Threatening Language

The message threatens account suspension and closure if immediate action is not taken.

Risk:

Fear-based language is a common social engineering technique used in phishing attacks.

Identified Red Flags

No.	Red Flag	Severity
1	Fake sender address	High
2	Suspicious URL	Critical
3	Urgency and pressure	High
4	Generic greeting	Medium
5	Threatening language	High
6	Request for immediate action	High

Why This Email Is Unsafe

This message attempts to manipulate the recipient into revealing sensitive information through deception. The attacker impersonates a trusted organization and uses urgency, fear, and a fraudulent website to obtain user credentials. If a victim enters their information, the attacker may gain unauthorized access to financial accounts and personal data.

Security Recommendations

1. Never click links from unexpected emails.
2. Verify the sender's email address carefully.
3. Access websites directly through their official URLs.
4. Enable Multi-Factor Authentication (MFA).
5. Report suspicious emails to the IT or Security Department.
6. Delete phishing messages after reporting them.

Phishing Detection Checklist

Security Check	Status	Notes
Sender email address verified	No	The sender's address appears suspicious and does not match the official company domain.
Official domain used in links	No	The URL does not belong to the legitimate website.
Suspicious link detected	Yes	The link may redirect users to a fake login page.
Urgent or pressure language found	Yes	The email demands immediate action within 24 hours.
Generic greeting used	Yes	The message starts with "Dear Customer" instead of the recipient's name.
Threatening language present	Yes	The email threatens account suspension or closure.
Request for sensitive information	Yes	The message attempts to obtain account credentials.
Safe to interact with	No	The email should be considered unsafe and potentially malicious.

Risk Level: High Risk

Conclusion

After analyzing the email, several phishing indicators were identified, including sender spoofing, a malicious URL, urgency tactics, generic greetings, and threatening language. These indicators strongly suggest that the message is a phishing attempt. Therefore, users should avoid interacting with the email and follow cybersecurity best practices to protect their personal and financial information.

Final Classification: Phishing Attack (Malicious Email)